

Agentic RAG

↳ Basic

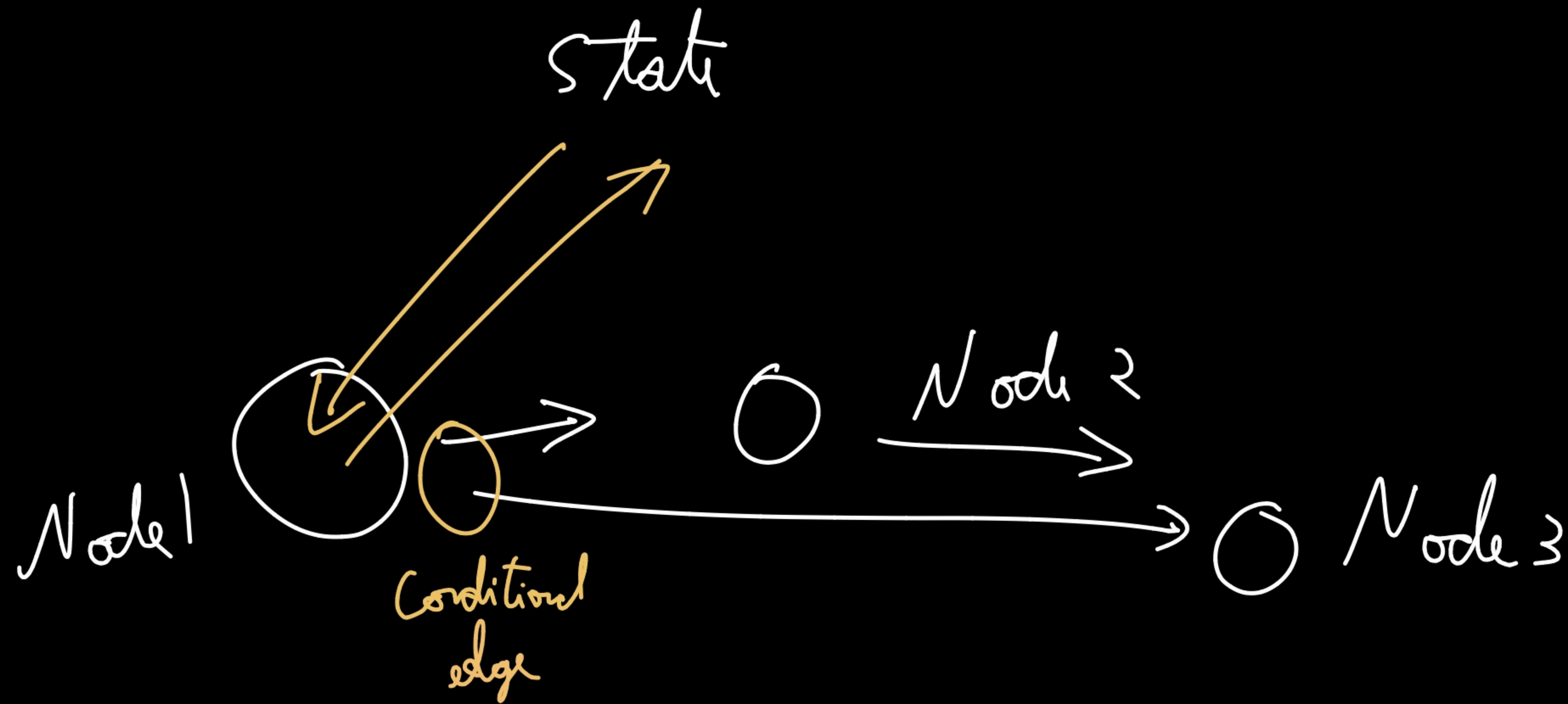
↳ Advanced

Offline & online —
understand design ✓

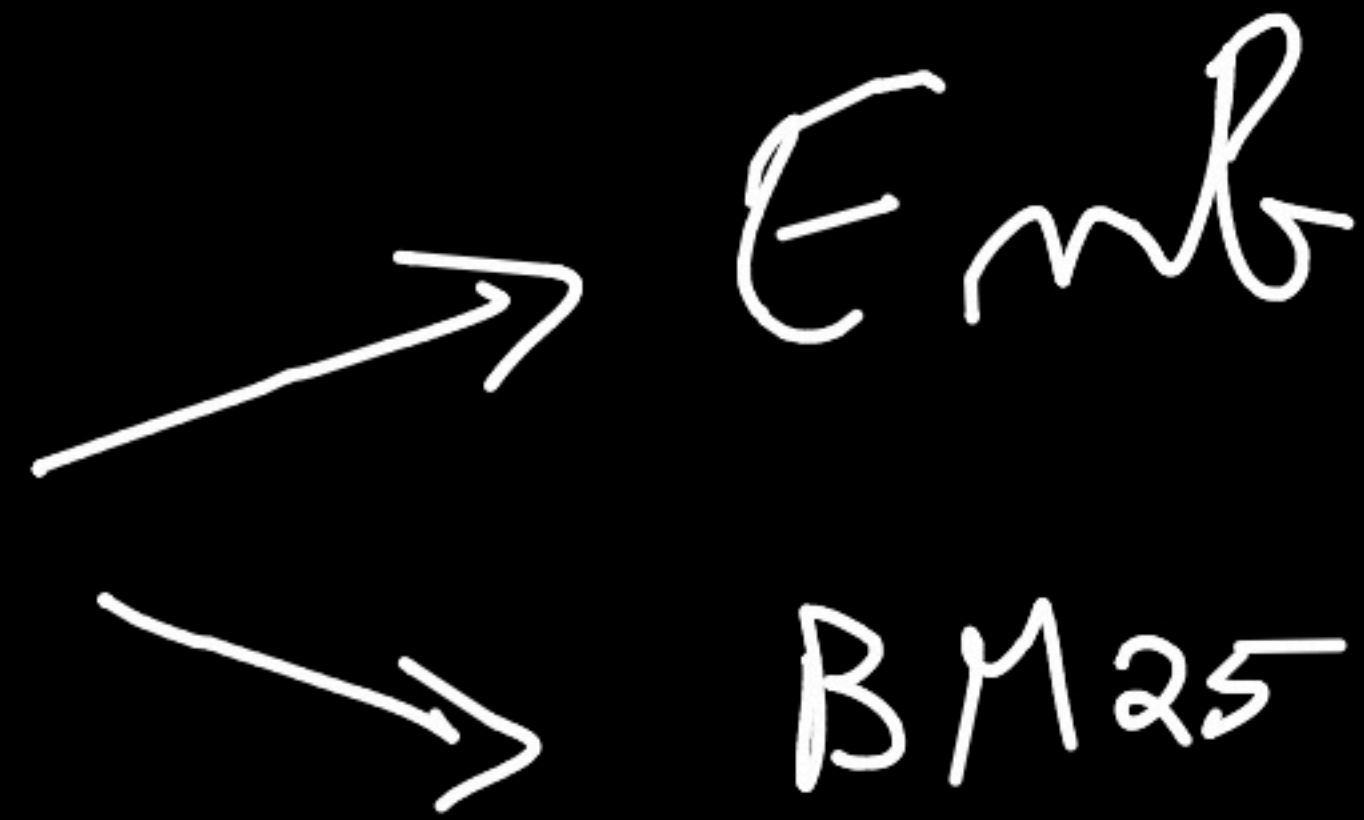
Improving Embedding model

Code walk through
minimize

↳ errors / failure



Offline



vectorstore (pk2)
BM25 (pk2)

query expansion/rewrite

chunking strategies

BM25

ReRank

Reasoning loops

Embedding
model



Domain

Specific data

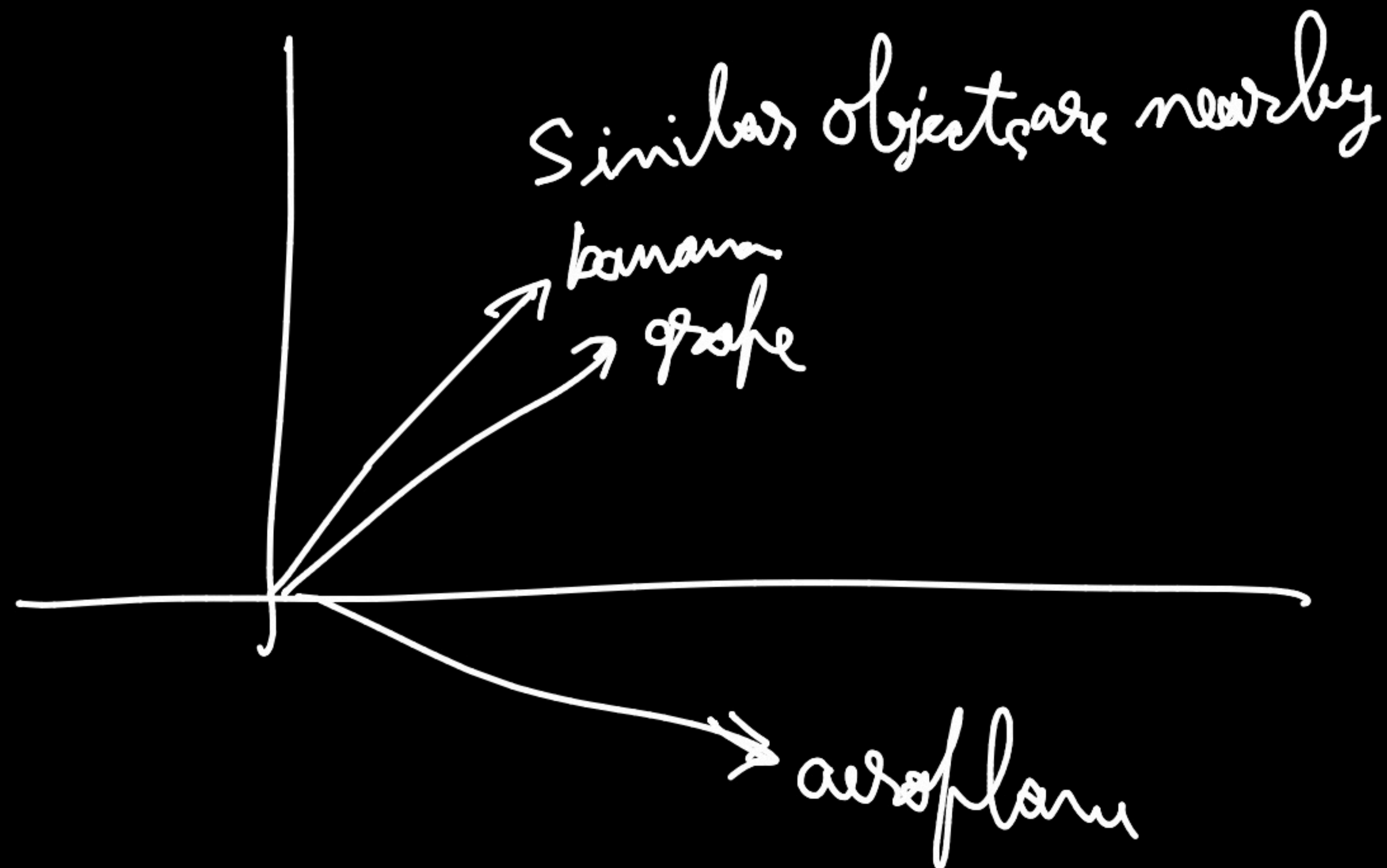
Update the
internal parameters
of the model

finetuning

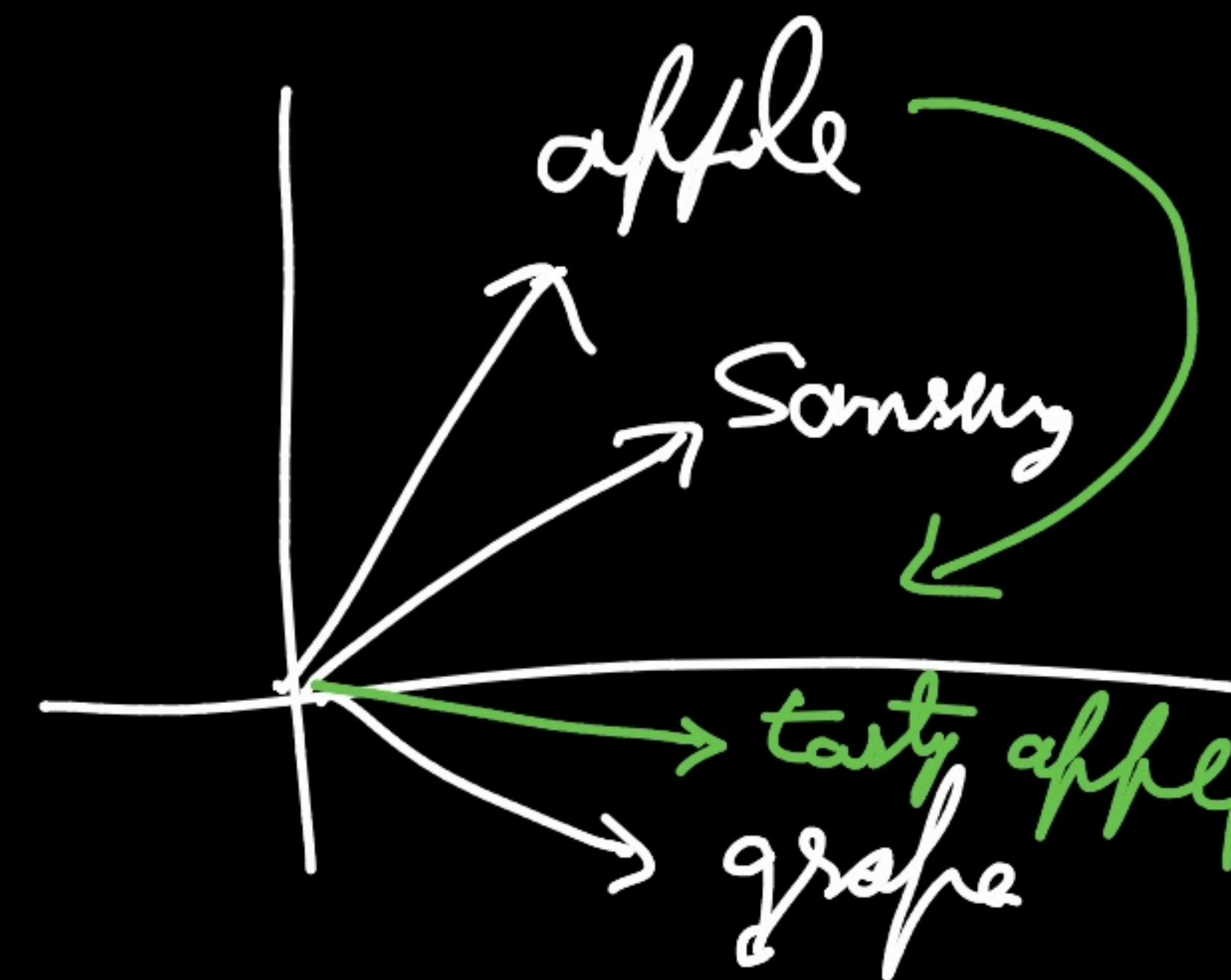
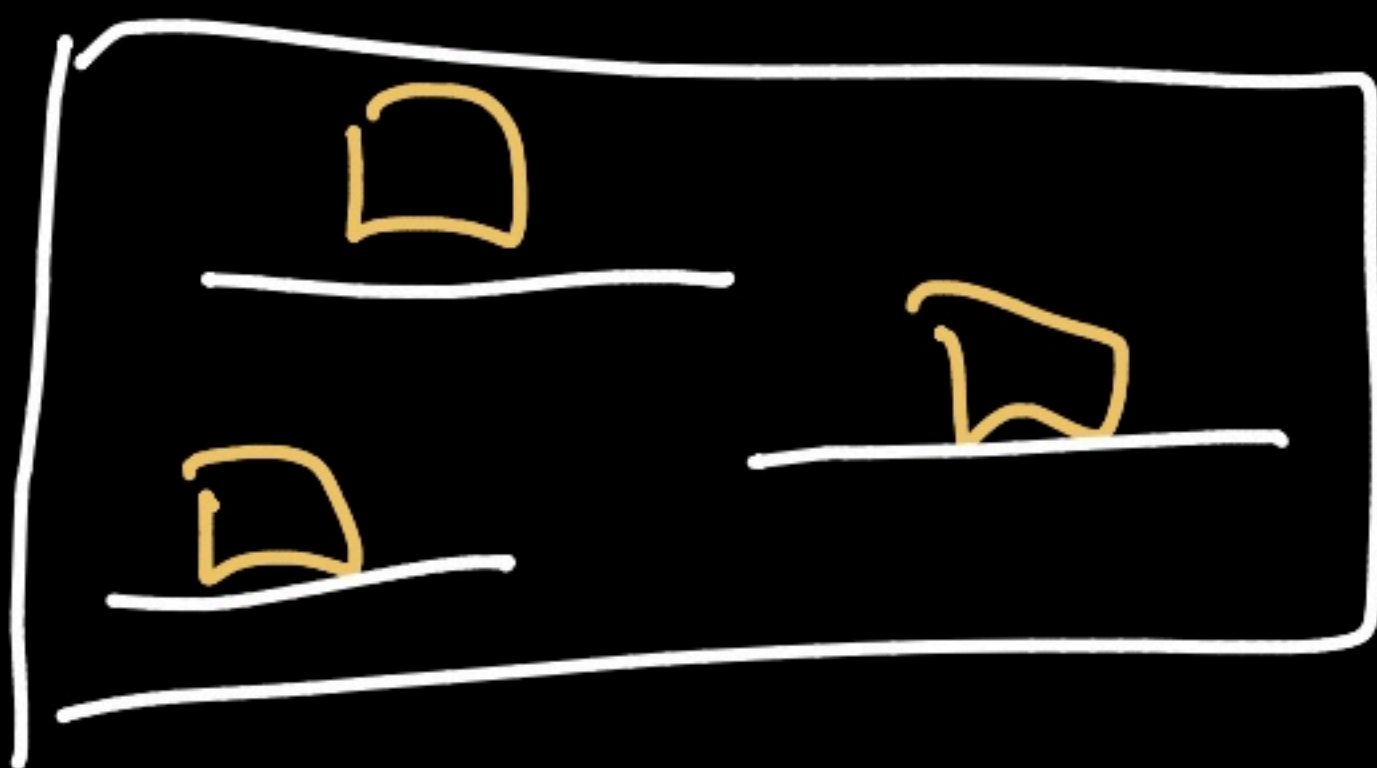
(Last resort)

Embedding model

Cosine similarity



Intuit

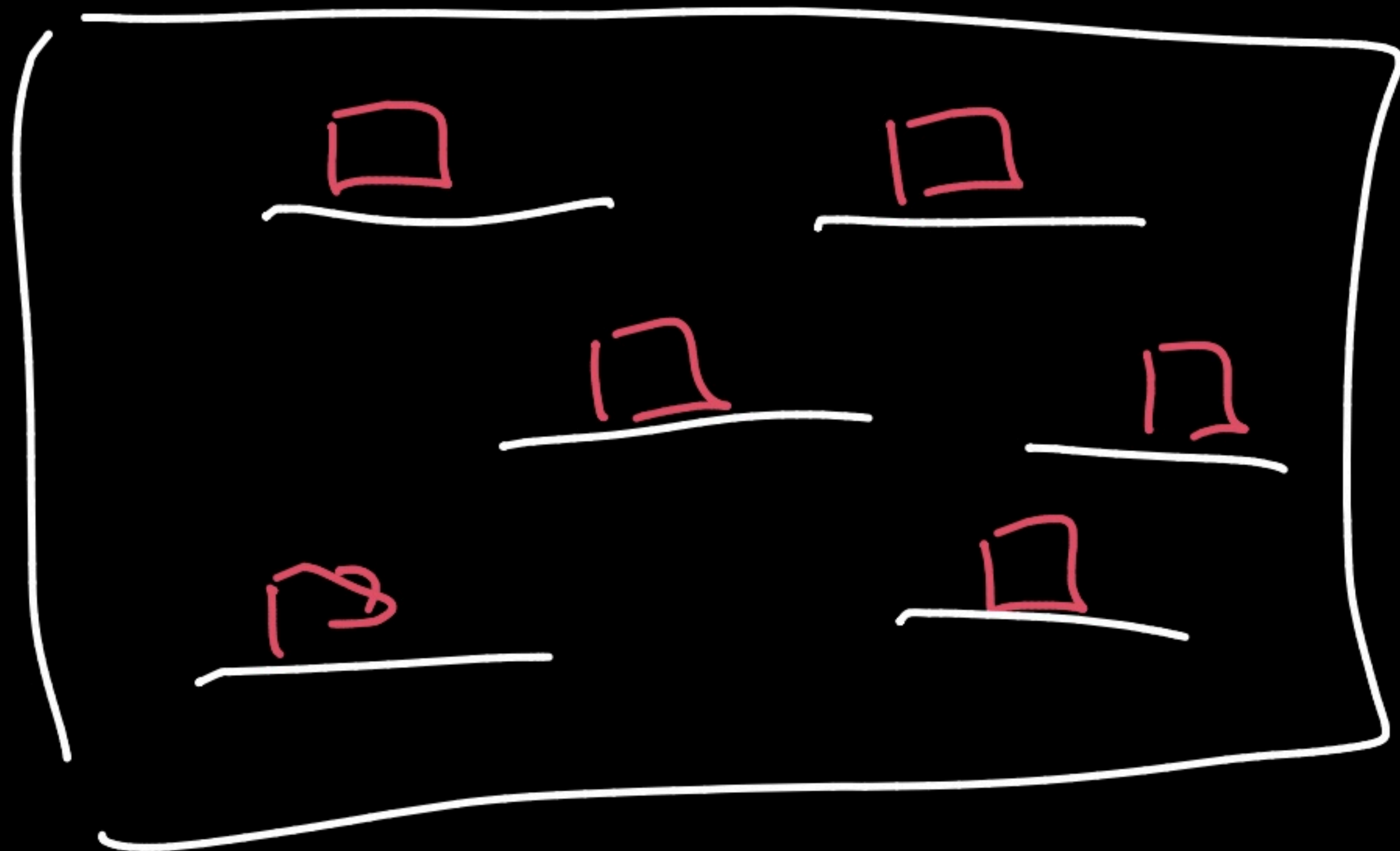


Anchor,
Carrot

search

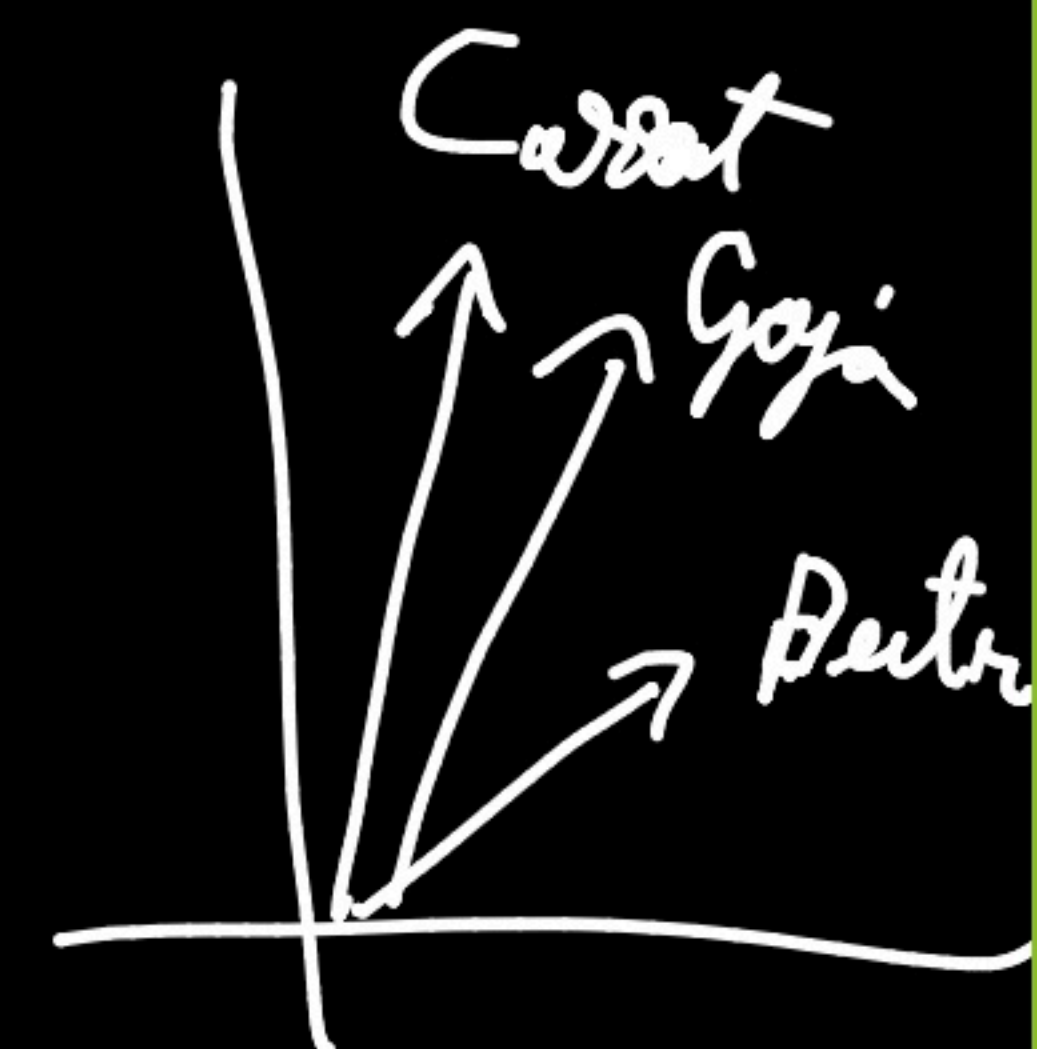
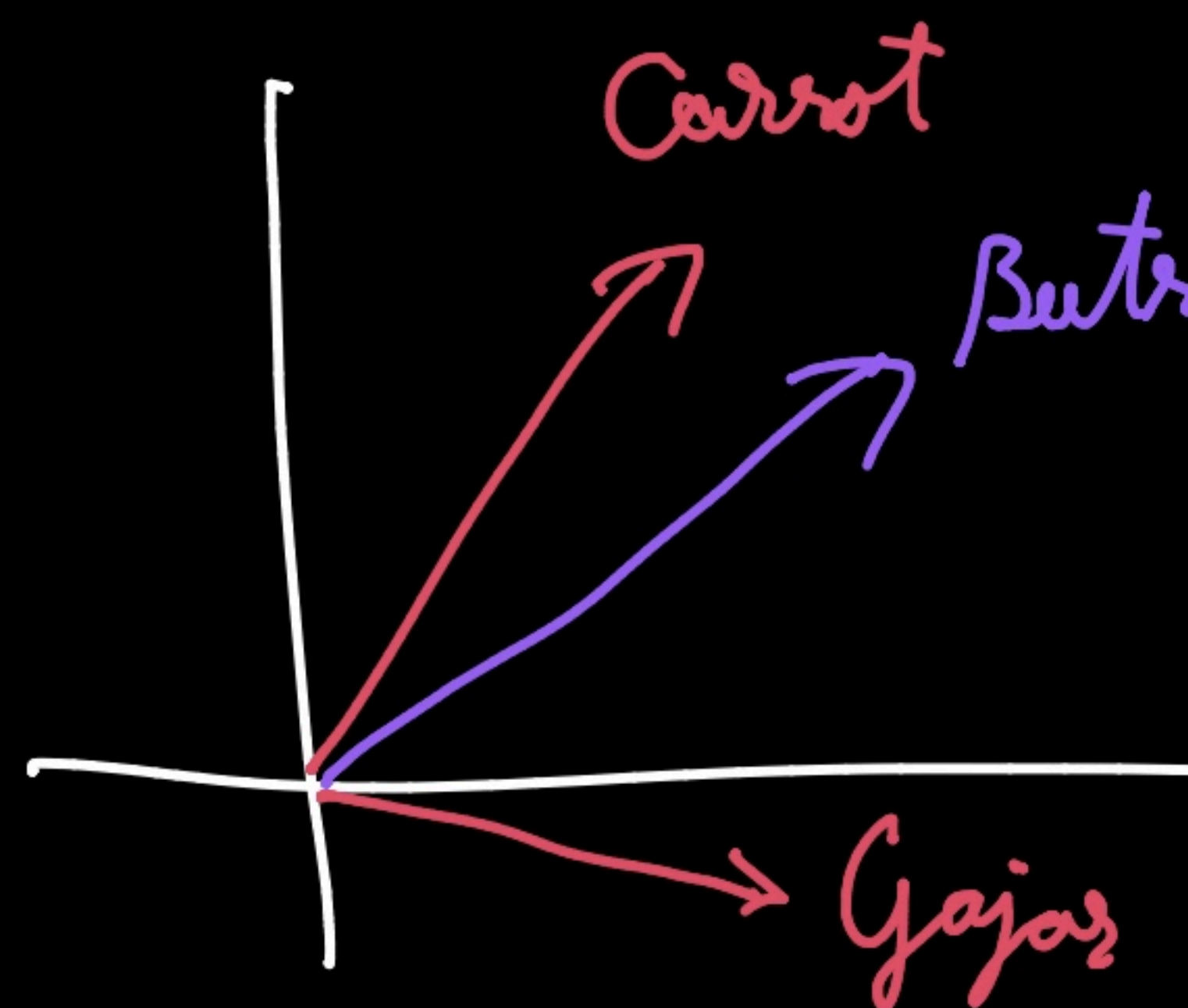
$w \rightarrow$ parameter

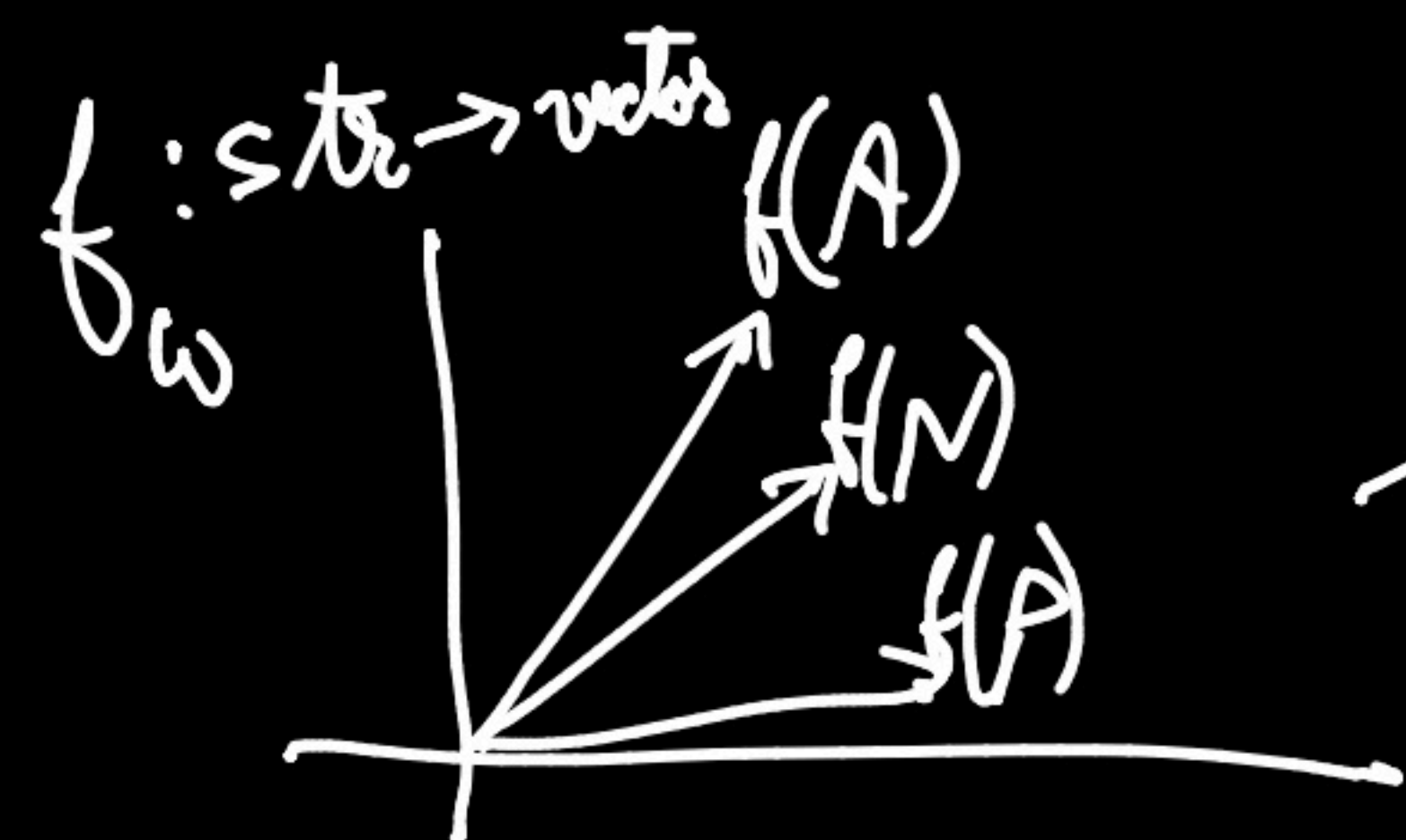
Carrot $f_w: \text{str} \rightarrow \mathbb{R}^m$



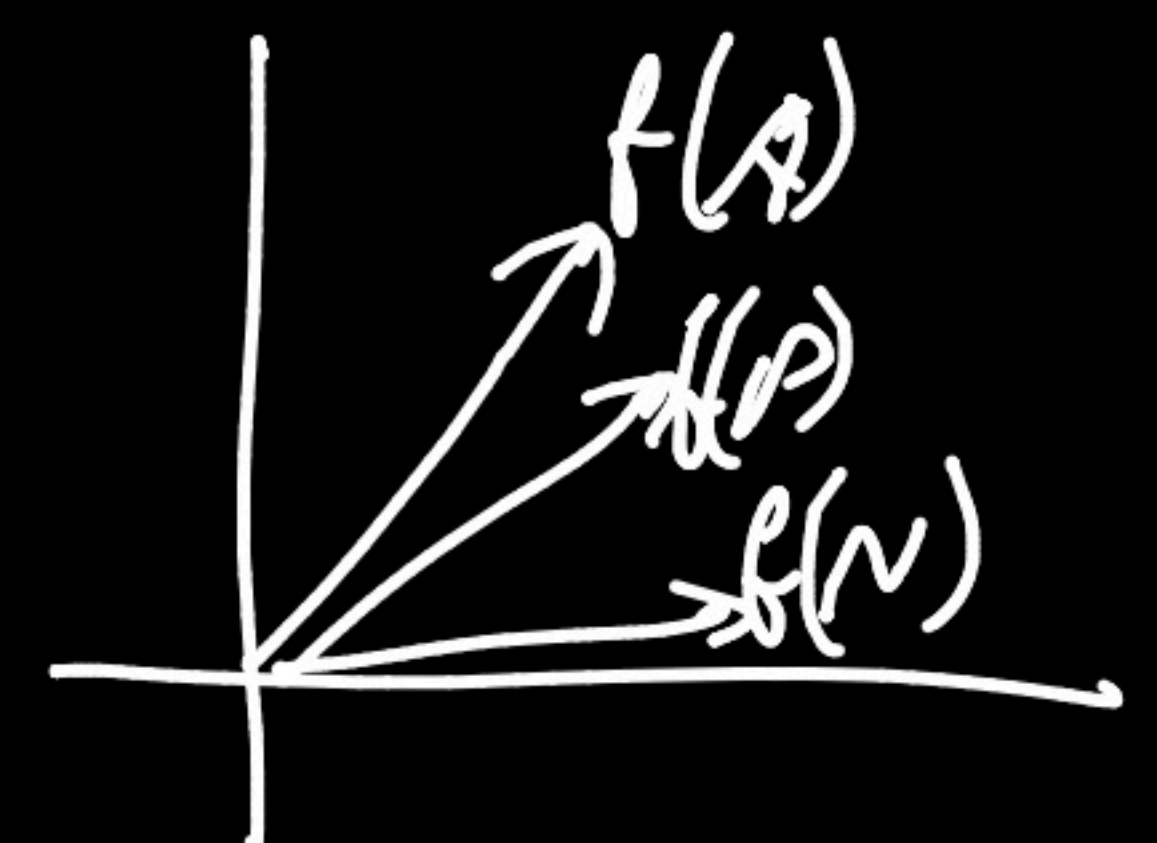
Gajar ^{Positive} (Hindi for carrot)

Beetroot ^{Negative}





finetuning



"Bad"

"good"

	A	P	N		Expertise Efforts
$i=0$	-	-	-	✓	
$i=1$	-	-	-	✓	
$\vdots$	-	-	-	X	
$i=m$	-	-	-	X	

$$\|f(A) - f(P)\|^2 + \alpha < \|f(A) - f(N)\|^2$$

$$\sum_{i=1}^m \max \left(\|f(A_i) - f(P_i)\|^2 - \|f(A_i) - f(N_i)\|^2 + \alpha, 0 \right)$$

53% 67% good
 ↓ ↓
 82% 85% finetuning
 ↓
 90%